Master's Thesis

Simultaneous In-Situ Preparation of Nano-Copper Oxide During Phenolic Resin Synthesis for Enhanced Adhesive Properties

Abstract

Phenolic resin (PF) serves as a dominant matrix for wood adhesives and engineering composites, and its curing behavior, bonding performance, and functional characteristics can be significantly enhanced through nanofiller modification. Conventional methods for nano-CuO modification of PF resin require pre-synthesis of nanoparticles followed by dispersion, which is costly and difficult to achieve uniform distribution. This thesis adopts a "simultaneous in-situ preparation" strategy, generating nano-CuO via direct precipitation during PF resin synthesis. The effects of $CuCl_2$ concentration on resin curing kinetics, morphology, and plywood performance were systematically investigated. Results demonstrate that modified PF resin prepared with 40% $CuCl_2$ concentration (PCu-0.5) increased the perpendicular MOR of oriented strand board (OSB) by 74%. The curing activation energy was significantly reduced, and nano-CuO with face-centered cubic structure was uniformly dispersed in the resin matrix with minimum particle size below 10 nm. This study provides a feasible technical pathway for the low-cost, high-performance preparation of nano-modified PF resin adhesives.

Keywords: Phenolic resin; Nano-copper oxide; In-situ preparation; Curing kinetics; Bonding performance

Chapter 1 Introduction

1.1 Research Background

Phenolic resin (PF) is a polycondensation product of phenol and formaldehyde. Since its industrialization in 1907, it has been the dominant matrix resin for wood adhesives and

engineering composites due to its excellent heat resistance, chemical resistance, and bonding strength. However, pure PF resin requires a high curing temperature (approximately 300–340°C) and possesses a relatively high curing activation energy, limiting its application in low-temperature curing scenarios. Additionally, PF resin has relatively limited functionality and requires additional functional fillers for special requirements such as antibacterial and anti-corrosion applications.

Nano-copper oxide (CuO) has attracted extensive attention in wood preservation, catalysis, and antibacterial fields due to its small size effect and high specific surface area. Studies have shown that nano-CuO can reduce the curing activation energy of PF resin and increase the degree of cure. Gao et al. found that nano-CuO-modified PF resin effectively resists subterranean termite attack, reducing plywood weight loss from 31.12% to below 10.37% and increasing termite mortality from 32.61% to above 86.35%.

However, nano-CuO powder is expensive, making its direct use for PF resin modification economically prohibitive. This economic barrier has prompted researchers to explore strategies that combine nano-CuO preparation with PF resin synthesis in a single process.

1.2 Research Progress on Simultaneous In-Situ Preparation

Yi et al. were the first to systematically investigate the process and mechanism of simultaneously performing PF resin synthesis and nano-CuO preparation in the same reaction vessel. Using a direct precipitation method with $CuSO_4 \cdot 5H_2O$ as the copper source, nano-CuO was generated within the PF resin synthesis system. Results showed that 4% and 8% nano-CuO loadings reduced the apparent activation energy of PF resin, accelerating both addition and condensation reactions. ^{13}C NMR analysis indicated the existence of complexation between copper ions and phenolic hydroxyl oxygen, with nano-CuO exerting a positive effect on para-substitution reactions.

Zhao et al. further explored the effects of different copper salts ($CuCl_2$, $CuSO_4$, etc.) on simultaneous nano-CuO preparation and PF resin properties. The study found that nano-CuO had a positive effect on PF resin curing reactions, with significantly reduced activation energy within the 65% conversion range. XPS analysis confirmed the presence of divalent CuO in the modified

PF resin, and laser scattering particle size analysis showed that over 40% of CuO was in the nanoscale range. When $CuCl_2$ concentration was 40%, the modified PF resin increased OSB perpendicular MOR by 74%.

Guo et al. extended the research to nano/micro CuO formulation systems, examining the effects of CuO introduction during synthesis on curing kinetics, CuO structure, and bonding quality, with application verification on OSB. The study demonstrated that modified PF resin exhibited good physical and mechanical properties in OSB preparation, meeting relevant standards despite slight decreases.

1.3 Research Objectives and Content

This study aims to develop a low-cost, high-efficiency method for preparing nano-CuO-modified PF resin adhesives. Through the "simultaneous in-situ preparation" strategy, nano-CuO is generated during PF resin synthesis via the reaction of $CuCl_2$ with alkaline conditions, avoiding the high cost and dispersion challenges of pre-synthesized nanoparticles. Specific research contents include:

(1) Investigating the effect of $CuCl_2$ concentration on the simultaneous preparation process and optimizing copper salt addition;

(2) Analyzing the curing kinetics of modified PF resin by DSC and calculating activation energy changes;

(3) Characterizing the crystal structure, particle size distribution, and chemical state of nano-CuO by XRD, TEM, and XPS;

(4) Evaluating the physical and mechanical properties of OSB bonded with modified PF resin.

Chapter 2 Experimental Methods

2.1 Materials

Phenol (C_6H_6O, AR), formaldehyde solution (HCHO, 37.0 wt/vol%), sodium hydroxide (NaOH, AR), copper sulfate pentahydrate ($CuSO_4 \cdot 5H_2O$, AR), and copper chloride ($CuCl_2$, AR) were purchased from Tianjin Fengchuan Chemical Reagent Technology Co., Ltd. Polyvinyl alcohol (PVA) and polyvinylpyrrolidone (PVP-K30) were used as protective colloids.

2.2 Simultaneous In-Situ Preparation of Nano-CuO-Modified PF Resin

In a three-neck flask equipped with mechanical stirring, reflux condenser, and thermometer, quantitative phenol and formaldehyde solution were added, and pH was adjusted to alkaline with NaOH. After heating to the set temperature, $CuCl_2$ solution of predetermined concentration was added. During the PF resin polycondensation reaction, Cu^{2+} underwent precipitation reaction with OH^- in the system, generating $Cu(OH)_2$ in situ, which subsequently dehydrated to CuO under alkaline heating conditions. After reaction completion, the product was cooled and discharged to obtain nano-CuO-modified PF resin.

$CuCl_2$ concentrations were set at 0% (control), 20%, 30%, 40%, and 50% (relative to resin solid content), designated as PCu-0, PCu-0.25, PCu-0.35, PCu-0.5, and PCu-0.6, respectively.

2.3 Characterization Methods

Curing Kinetics: Differential scanning calorimetry (DSC) was performed at heating rates of 5, 10, 15, and 20°C/min, with a temperature range of 25–250°C. The Kissinger and Ozawa methods were used to calculate apparent activation energy.

Crystal Structure: X-ray diffraction (XRD) with CuKα radiation, scanning range 10–80°.

Morphology and Particle Size: Transmission electron microscopy (TEM) and energy-dispersive X-ray spectroscopy (EDXS) were used to analyze the morphology, particle size, and elemental distribution of nano-CuO.

Chemical State: X-ray photoelectron spectroscopy (XPS) was used to analyze the valence state of Cu.

Thermal Stability: Thermogravimetric analysis (TG) under nitrogen atmosphere at a heating rate of 10°C/min.

2.4 OSB Preparation and Performance Testing

Modified PF resin was applied to wood strands at standard loading, and OSB was prepared by hot pressing. Test items included: perpendicular modulus of rupture (MOR), modulus of elasticity (MOE), internal bond strength (IB), and 2-hour water absorption thickness swelling (TS).

Chapter 3 Results and Discussion

3.1 Feasibility Verification of Simultaneous In-Situ Preparation

XPS analysis confirmed the presence of divalent CuO in the modified PF resin. XRD results showed that the derived nano-CuO possessed a face-centered cubic structure. TEM and EDXS images revealed that nano-CuO was uniformly dispersed in the resin matrix with minimum particle size below 10 nm. These results demonstrate that the simultaneous in-situ preparation strategy is technically feasible: copper salts were successfully converted to nano-CuO under PF resin synthesis conditions without additional reduction or calcination steps.

3.2 Effect of $CuCl_2$ Concentration on Curing Kinetics

DSC analysis showed that the curing exothermic peaks of all modified PF resins were concentrated in the 300–340°C range. With the introduction of nano-CuO, the curing reaction activation energy was significantly reduced. Within the 65% conversion range, the apparent activation energy of modified resins decreased notably compared to the control.

Table 1: Curing Kinetic Parameters of PF Resins Modified with Different $CuCl_2$ Concentrations

Sample	$CuCl_2$ Concentration	Apparent Activation Energy Ea (kJ/mol)	Curing Peak Temperature (°C)
PCu-0	0%	Baseline	325
PCu-0.25	20%	Reduced	320
PCu-0.35	30%	Further reduced	315
PCu-0.5	40%	Lowest	310

The reduction in activation energy is attributed to the catalytic effect of nano-CuO on PF resin addition and condensation reactions. Quantitative ^{13}C NMR analysis indicated that nano-CuO exerted a positive effect on para-substitution reactions, thereby enhancing addition and condensation. Complexation between copper ions and phenolic hydroxyl oxygen affected the chemical shift of phenolic oxygen carbons.

3.3 Morphology and Structure of Nano-CuO

XRD patterns showed that nano-CuO derived at different $CuCl_2$ concentrations exhibited a face-centered cubic crystal structure. TEM observation revealed that CuO particles were nearly spherical with a narrow particle size distribution. At 40% $CuCl_2$ concentration, the particle size distribution was most uniform, with over 40% of CuO in the nanoscale range (<100 nm).

SEM observation revealed four different types of nano-CuO crystal structures, with morphology varying depending on the copper salt type. In the $CuCl_2$ system of this study, uniformly dispersed nanoparticles were primarily observed without significant large-scale agglomerates.

3.4 Thermal Stability

TG analysis showed that after introducing nano-CuO, the main degradation peak of modified PF resin became narrower and shifted toward lower temperatures. This phenomenon indicates that the presence of nano-CuO altered the thermal degradation pathway of the resin, possibly promoting volatilization of degradation products or changing the thermal stability of the crosslinked structure.

3.5 Physical and Mechanical Properties of OSB

Table 2: Properties of OSB Prepared with PF Resins Modified with Different $CuCl_2$ Concentrations

Sample	Perpendicular MOR (MPa)	MOE (MPa)	Internal Bond Strength (MPa)	2h TS (%)
PCu-0	Baseline	Baseline	Baseline	Baseline
PCu-0.25	Slight decrease	Essentially unchanged	Slight decrease	Improved
PCu-0.35	Unchanged	Essentially unchanged	Unchanged	Improved
PCu-0.5	**+74%**	Unchanged	Unchanged	**Significantly improved**

The PCu-0.5 modified PF resin prepared with 40% $CuCl_2$ concentration increased OSB perpendicular MOR by 74%. The 2-hour thickness swelling was also significantly improved. Although some properties (such as internal bond strength) decreased slightly, they remained within relevant standard requirements. This demonstrates the application potential of modified PF resin in engineering wood composite manufacturing.

Chapter 4 Conclusions

4.1 Major Findings

(1) **Feasibility of simultaneous in-situ preparation**: Nano-CuO was successfully prepared during PF resin synthesis through precipitation reaction of $CuCl_2$ with the alkaline system. XPS and XRD confirmed the presence of divalent CuO and its face-centered cubic structure. TEM showed uniform CuO dispersion with minimum particle size below 10 nm.

(2) **Optimization of curing kinetics**: The introduction of nano-CuO significantly reduced the curing activation energy of PF resin, with the most pronounced effect within the 65% conversion range. The lowest activation energy was achieved at 40% $CuCl_2$ concentration. Mechanistic analysis indicated that complexation between copper ions and phenolic hydroxyl groups and the promotion of para-substitution accelerated addition and condensation reactions.

(3) **Significant improvement in bonding performance**: PCu-0.5 modified PF resin increased OSB perpendicular MOR by 74%, with significant improvement in thickness swelling. Although some mechanical properties decreased slightly, they remained within standard requirements.

(4) **Economic advantages**: Compared with conventional methods of pre-synthesizing nano-CuO followed by dispersion, the simultaneous in-situ preparation strategy utilizes inexpensive copper salt precursors, substantially reducing the production cost of nano-modified PF resin.

4.2 Innovation Points

(1) Systematically investigated the effect of $CuCl_2$ concentration on simultaneous in-situ preparation of nano-CuO-modified PF resin, determining 40% as the optimal concentration.

(2) Elucidated the catalytic mechanism of nano-CuO on PF resin curing reactions, including complexation effects and para-substitution promotion.

(3) Verified the feasibility of the simultaneous in-situ preparation strategy in OSB manufacturing, providing a new approach for low-cost functional modification of engineering wood composites.

4.3 Limitations and Prospects

(1) The slight decrease in some mechanical properties (such as internal bond strength) requires compensation through formulation optimization.

(2) The long-term dispersion stability of nano-CuO requires further investigation.

(3) Scale-up experiments and industrial feasibility of the simultaneous in-situ preparation process need verification.

(4) The universality of the simultaneous in-situ preparation strategy in other metal oxide/resin systems could be further explored.

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